



# Used-Vehicles Sales Analysis

For BUDT730 Final Project

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# Agenda

01

## **Business Problem**

- Business Problem
- Business Questions

02

## **Utilized Data**

- Utilized Data
- Data Cleaning Process

03

## **Statistical Analysis**

- Statistical Analysis for Individual Business Questions

04

## **Conclusion**

- Final Analysis and Business Values
- Limitations
- Next Steps

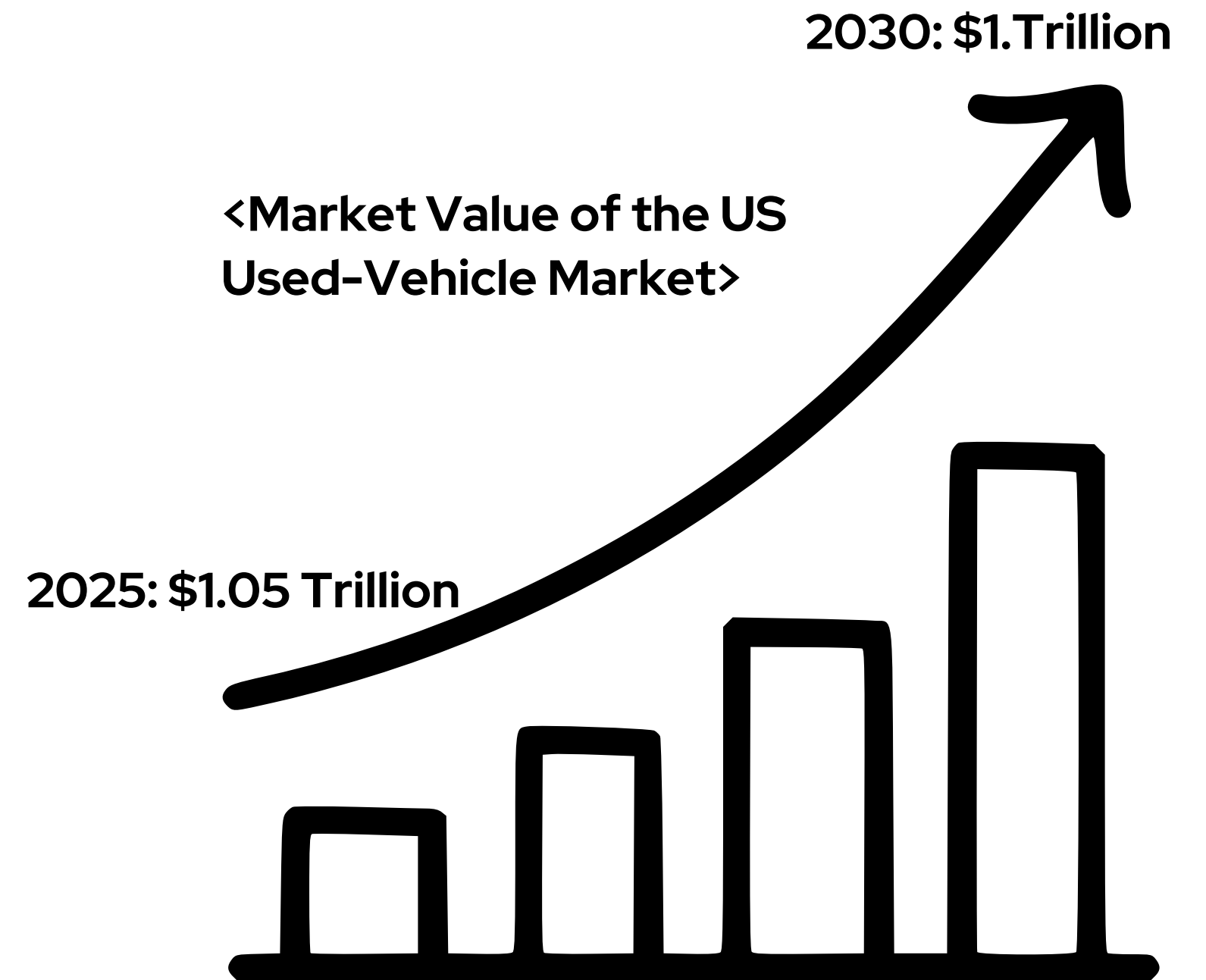
# Business Problem

- Business Problem
- Business Questions

# Business Problem

## ➡ Why Used-Vehicle Pricing Matters

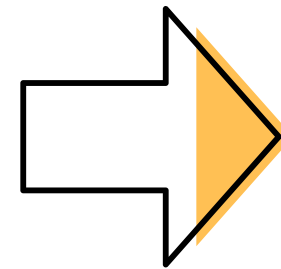
- U.S. used-vehicle market around \$1.05T in 2025, projected to \$1.20T by 2030.
- Large, competitive, and very sensitive to vehicle features, location, and timing
- Yet pricing still relies on rules of thumb and generic guides
- Not clear which specific attributes actually drive resale value



# Business Problem

## The Pain Points

- Pricing is unclear and non-uniform across the U.S.
- Sellers risk:
  - Underpricing → lost profit
  - Overpricing → slow inventory, higher holding costs
- Buyers face:
  - Confusing, inconsistent prices for similar cars



## Main Business Problem:

“Which factors most strongly influence the resale value of a used vehicle in the United States?”

## Needs

- Data driven analysis of what determines the resale value of used vehicles



# Business Questions

## **Main Business Problem:**

**"Which factors most strongly influence the resale value of a used vehicle in the United States?"**

01

How do resale value vary across states?

02

What is the distribution of the difference in the resale value of the car?

03

Which car brand holds value best in the used market?

04

What features are important to contribute to deciding the selling price of a used car?

# Utilized Data

- Utilized Data
- Data Cleaning Process

# Utilized Data and Cleaning

- **Data Source**

- Kaggle "Vehicle Sales and Market Trends" dataset.
- Contains over 550,000 transactions across the U.S. over an eight month period.

- **Standardizing Categorical Fields**

- Several categorical variables such as Make, Model, Body, Color, etc. contained missing values, inconsistent casing, or placeholder symbols.

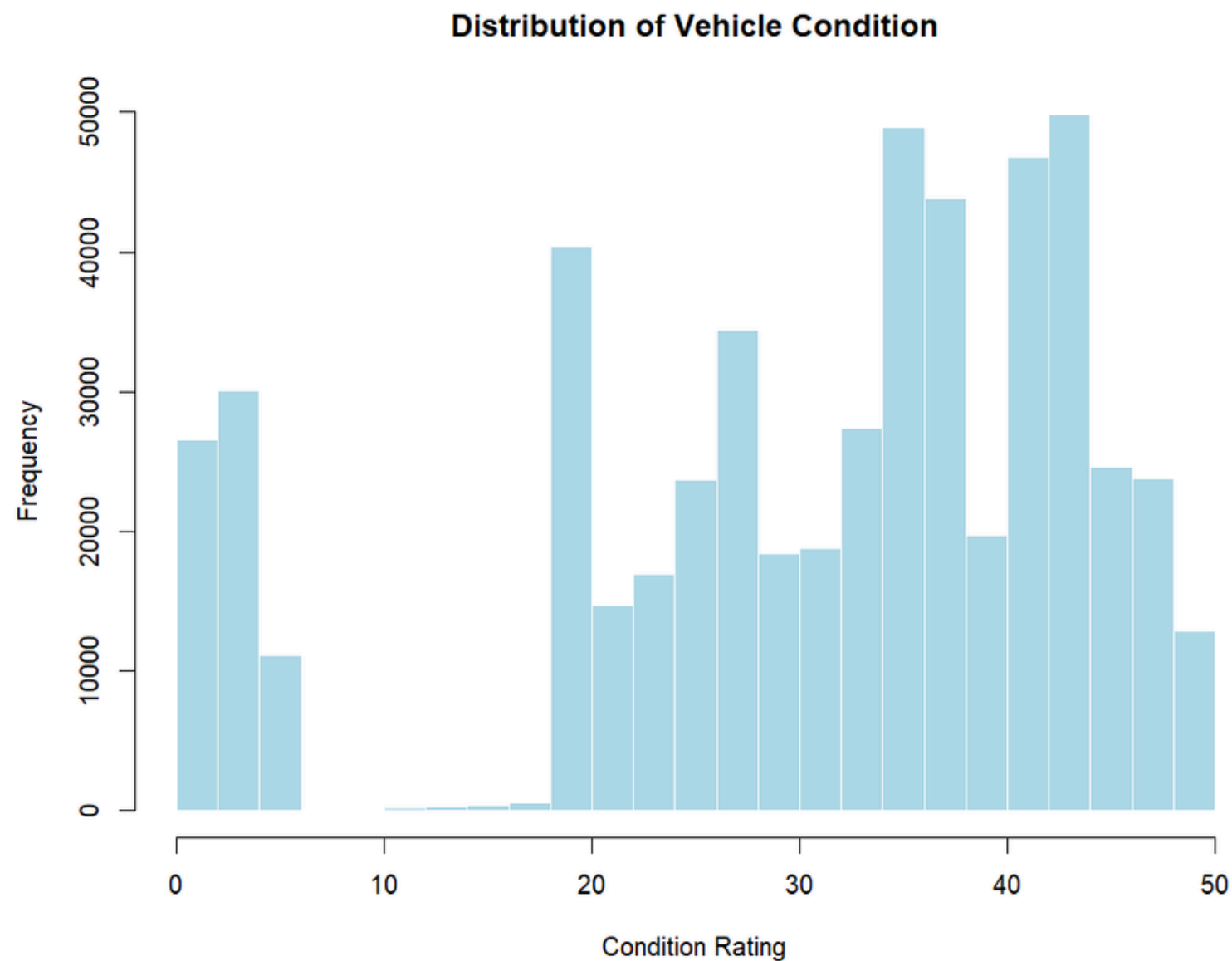
- **Managing Duplicate VINs**

- We found 8,000+ duplicate VINs. These represent the same car being sold multiple times.
- To avoid double counting, only the first recorded sale of each VIN was kept.
- A secondary dataset was created containing repeat sales for future analysis on multiple sale depreciation patterns.



# Utilized Data and Cleaning

## Data Cleaning



- Vehicle condition (0 - 50 scale) is a key predictor of price, but 11,000+ values were **missing**.
- The existing condition scores were **left-skewed** with many lower-quality vehicles.
- Since the distribution is skewed, we used **median** to impute missing values (more robust than mean).

# Statistical Analysis

- How do resale value vary across states?
- What is the distribution of the difference in the resale value of the car?
- Which car brand holds value best in the used market?
- What features are important to contribute to deciding the selling price of a used car?

# How do resale value vary across states?

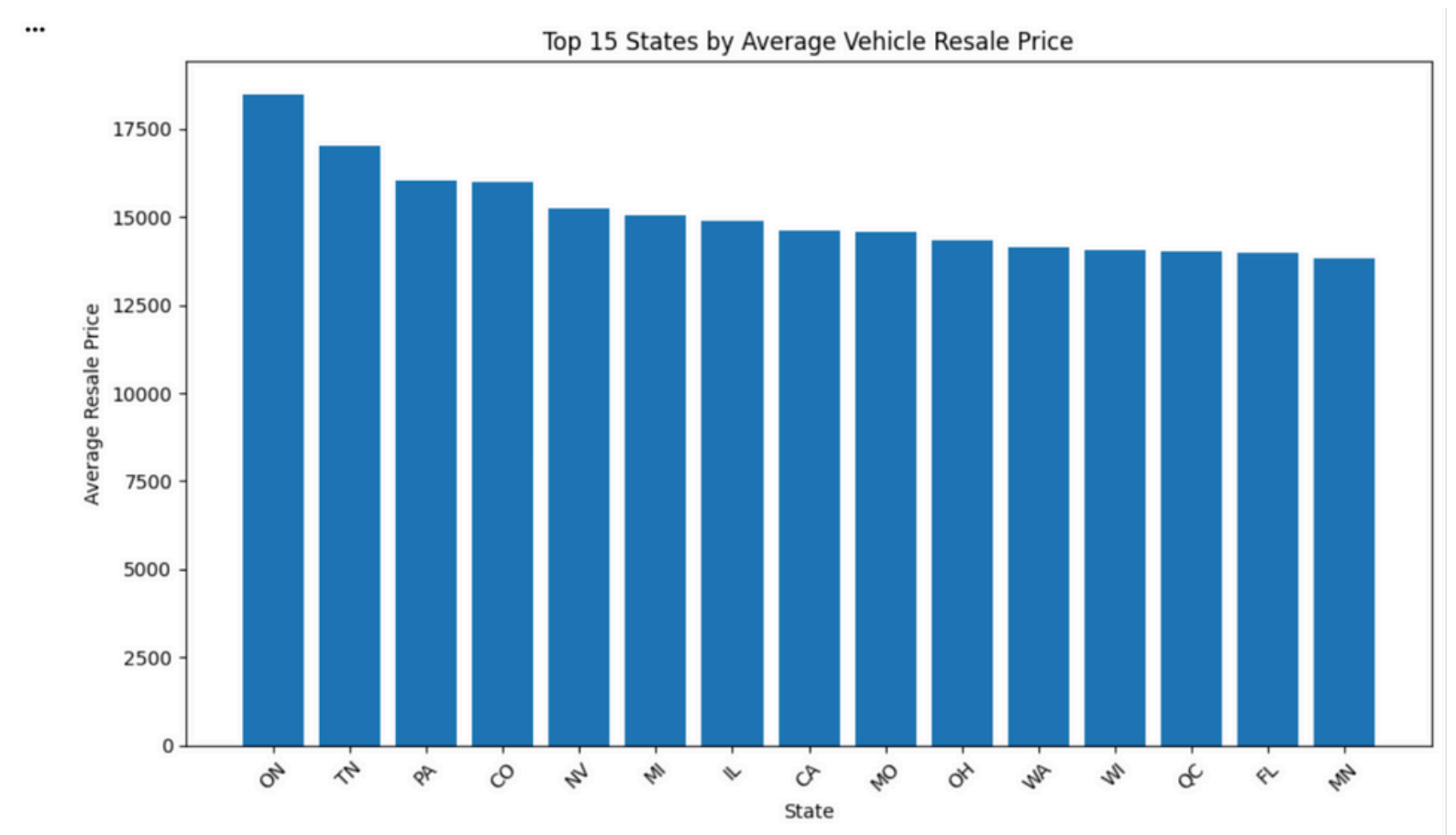
## Data Cleaning Process

- Standardized state codes
- kept only valid U.S. states
- Grouped transactions by **state**

## Calculated for each state

- Mean and median selling price
- Standard deviation and percentiles

## Identified Top 15 by Average Resale Price



# How do resale value vary across states?

## State-Level Results & Insights

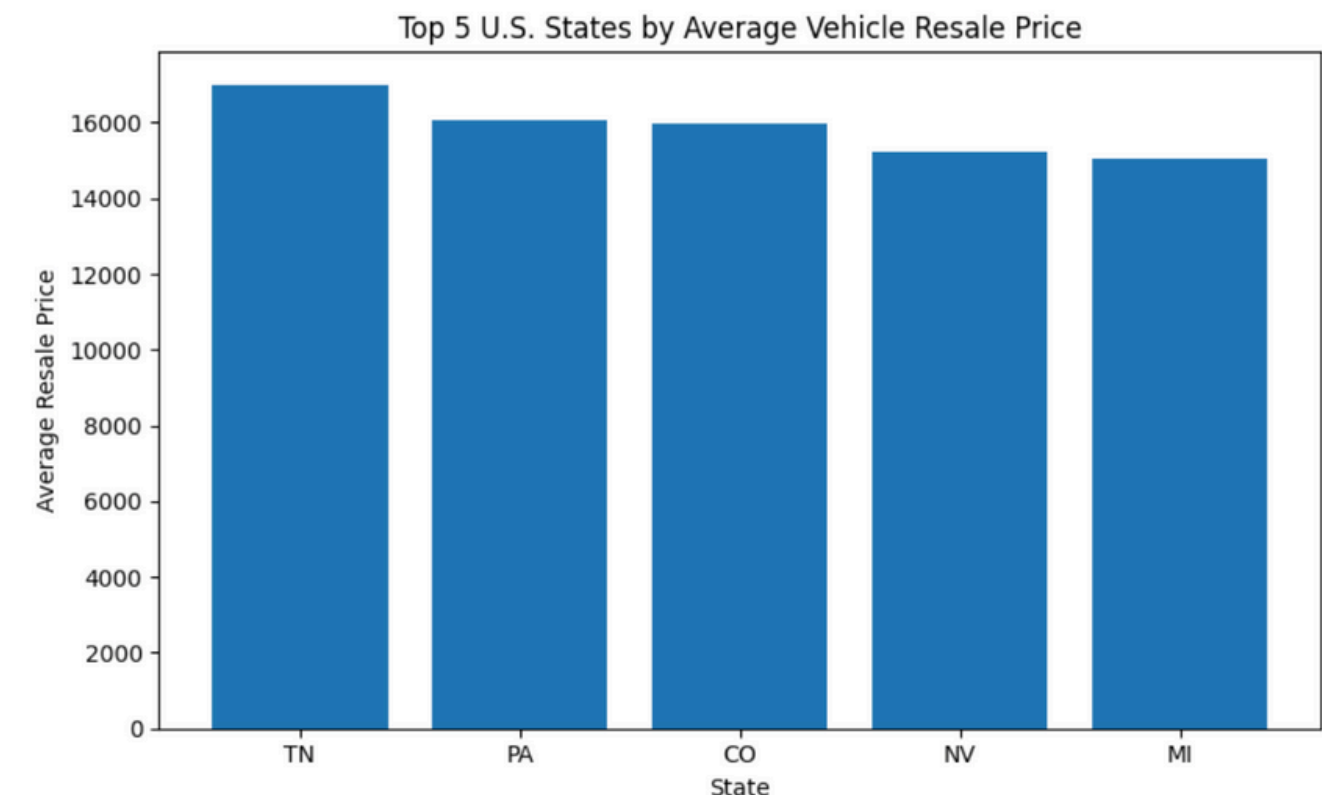
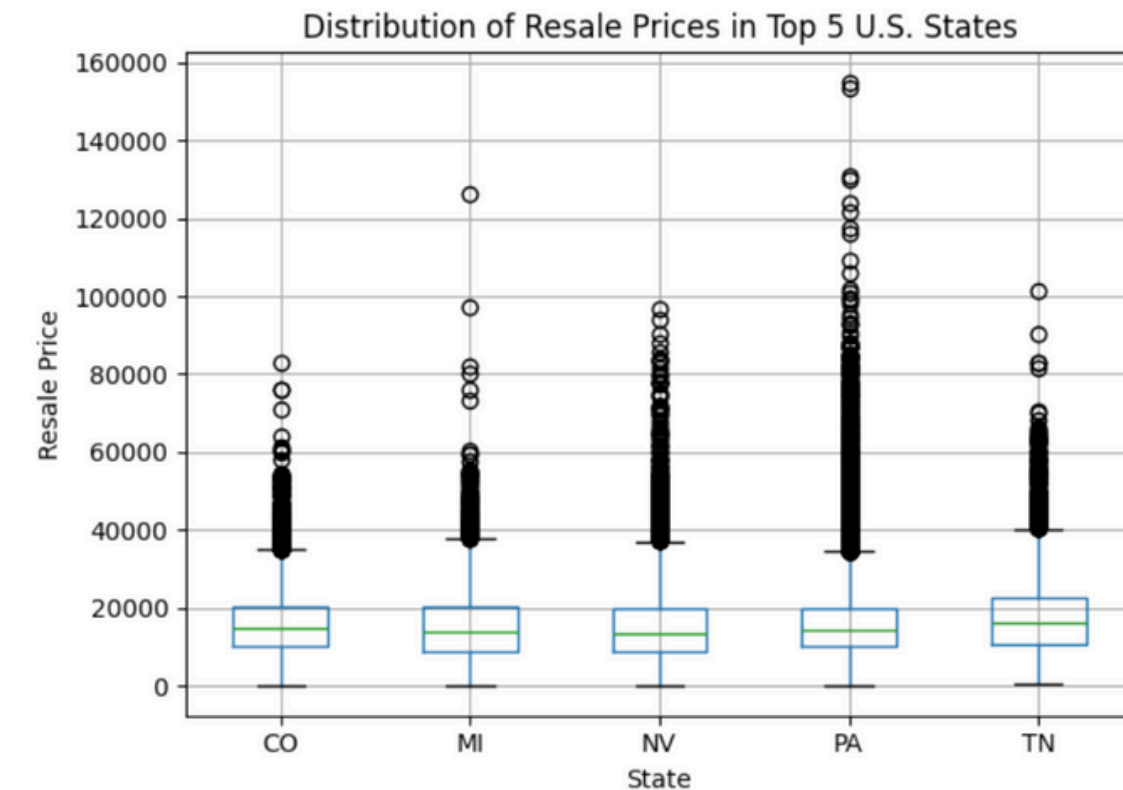
- Top states by average resale price:
  - Tennessee, Pennsylvania, Colorado, Nevada, Michigan
- Box plots show:
  - Higher medians and interquartile ranges, not just extreme outliers

## Takeaway

- Location (States) is a factor that effects the resale values of the vehicles

## Business impact

- Sellers can consider routing inventory to high-value states
- Buyers can look for lower-priced markets when feasible

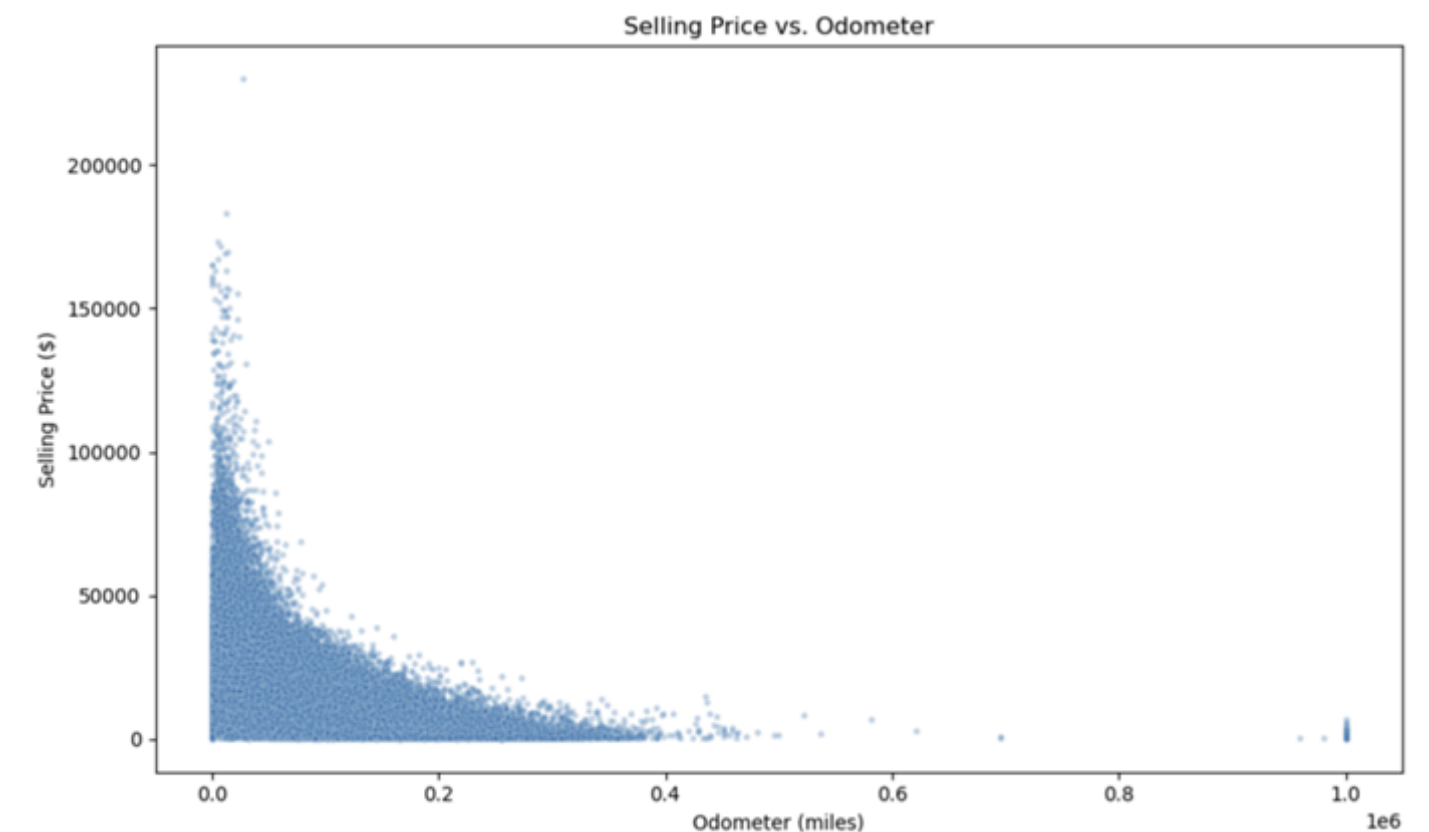


# Which car brand holds value best in the used market?

- **Odometer readings** are one of the strongest predictors of resale value
- The goal is to compare how quickly different brands lose value as mileage increases
- Key Challenge:
  - Different brands start at different price levels → raw prices are not comparable.
  - Example: BMW vs Toyota

## Why use log (selling price)?

- Raw relationship between selling price and mileage is non-linear
- Log Transform makes the relationship more linear
- Allows the coefficient to be interpreted as % value loss per mile
- Makes it fair to compare brands





# Which car brand holds value best in the used market?

## Regression Model

$$\ln(\text{SellingPrice}) = \beta_0 + \beta_1(\text{Mileage})$$

Where:

Dependent variable: log (selling price)

Predictor: mileage

$\beta_1$ : % value lost per mile

Results expressed as % lost per 10,000 miles

## Example: Toyota

- $\beta_1 = -9.4361 \times 10^{-6} (p < 0.001)$ 
  - => 9% value lost per 10,000 miles
- $R^2 = 0.572$  (strong for a single-variable model)

Brand: Toyota  
Dep. Variable: log\_price  
No. Observations: 39111  
R-squared: 0.572

Coefficients:

|          | coef              | std_err          | t_value   | p_value    |
|----------|-------------------|------------------|-----------|------------|
| const    | 9.9054160510e+00  | 3.8566043452e-03 | 2568.4294 | 0.0000e+00 |
| odometer | -9.4358195791e-06 | 4.1271531198e-08 | -228.6278 | 0.0000e+00 |

# Which car brand holds value best in the used market?

## Results across all brands

- Highest value retention:
  - Ram, Hummer, Toyota, Chevrolet
- Fastest depreciation:
  - BMW, Porsche, Maserati, Jaguar
  - Jaguar loses ~25% per 10,000 miles
- Strong pattern:
  - Durable/affordable brands → slow depreciation
  - Luxury brands → steep depreciation





# Which car brand holds value best in the used market?

## **Conclusion: which brands hold value best**

- Mileage based % depreciation creates a fair comparison across brands
- Value retention varies a lot between brands
- Buyers can identify which brands hold long-term value
- Sellers and dealerships can price more accurately
- Brands known for durability and affordability retain value, while luxury brands lose value quickly as mileage increases.

# What is the distribution of the difference in the resale value of the car ?

Statistical information of the dataset before outlier removal

|       | year          | condition     | odometer      | mmr           | sellingprice  | salesDifference |
|-------|---------------|---------------|---------------|---------------|---------------|-----------------|
| count | 536433.000000 | 536433.000000 | 536433.000000 | 536433.000000 | 536433.000000 | 536433.000000   |
| mean  | 2010.146262   | 30.888961     | 67253.802443  | 13866.636560  | 13707.906262  | -158.73029      |
| std   | 3.893405      | 13.256635     | 52832.770737  | 9610.701369   | 9679.060669   | 1741.06718      |
| min   | 1990.000000   | 1.000000      | 1.000000      | 25.000000     | 1.000000      | -87750.00000    |
| 25%   | 2008.000000   | 24.000000     | 27974.000000  | 7325.000000   | 7100.000000   | -800.00000      |
| 50%   | 2012.000000   | 35.000000     | 50968.000000  | 12350.000000  | 12250.000000  | -50.00000       |
| 75%   | 2013.000000   | 41.000000     | 97608.000000  | 18400.000000  | 18300.000000  | 650.00000       |
| max   | 2015.000000   | 49.000000     | 999999.000000 | 182000.000000 | 230000.000000 | 207200.00000    |

- Vehicles sold for an average of \$158.90 below their MMR value.
- Price variation (standard deviation) of approximately \$1,741 indicates significant spread.
- Wide variation suggests presence of extreme outliers in the dataset

Statistical information of the dataset after outlier removal

|       | year          | condition     | odometer      | mmr           | sellingprice  | salesDifference |
|-------|---------------|---------------|---------------|---------------|---------------|-----------------|
| count | 503110.000000 | 503110.000000 | 503110.000000 | 503110.000000 | 503110.000000 | 503110.000000   |
| mean  | 2010.088126   | 31.214583     | 68227.995442  | 13426.114070  | 13376.420320  | -49.915855      |
| std   | 3.931794      | 12.997754     | 52906.964345  | 9059.822551   | 9124.905015   | 1068.442540     |
| min   | 1990.000000   | 1.000000      | 1.000000      | 25.000000     | 1.000000      | -2700.00000     |
| 25%   | 2008.000000   | 25.000000     | 28652.000000  | 7000.000000   | 7000.000000   | -700.00000      |
| 50%   | 2012.000000   | 35.000000     | 51958.500000  | 12150.000000  | 12100.000000  | -25.00000       |
| 75%   | 2013.000000   | 42.000000     | 98990.000000  | 17950.000000  | 18000.000000  | 600.00000       |
| max   | 2015.000000   | 49.000000     | 999999.000000 | 182000.000000 | 183000.000000 | 2500.00000      |

- Vehicles sold for an average of \$49.90 below their MMR value—nearly at market price.
- Reduced standard deviation of approximately \$1,068 shows much tighter price clustering.
- Successful outlier removal resulted in significantly less variability in the dataset

# What is the distribution of the difference in the resale value of the car ?

## **Overall Result**

- Most vehicles sell close to their MMR value after data cleaning
- The distribution shifted closer to zero, indicating better alignment with market estimates
- Removing outliers reduced variability by ~39%, revealing the true underlying pattern.
- The slight negative mean suggests vehicles typically sell marginally below MMR on average.

# What features are important to contribute to deciding the selling price of a used car?

**Model A: Building a model using raw model and comparing p-value of different features to check for statistically significance**

```
Call:
lm(formula = sellingprice ~ year + make + model + trim + body +
    transmission + state + condition + odometer + color + interior,
    data = sample_df)

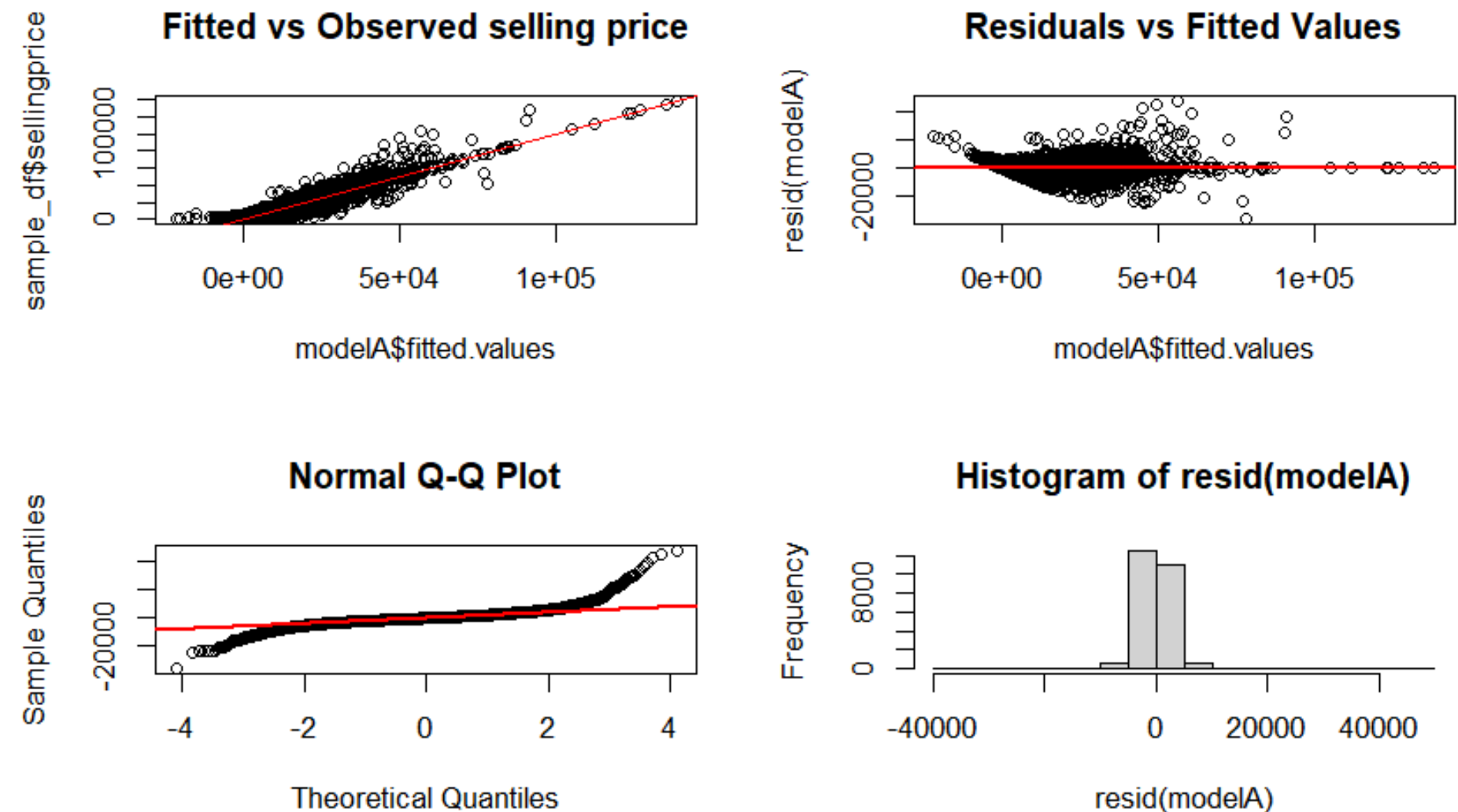
Coefficients: (121 not defined because of singularities)
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.920e+04  4.503e+03   4.264 2.02e-05 ***
year1991      7.979e+02  3.710e+03   0.215 0.829747
.
makeBentley   4.516e+04  3.754e+03  12.030 < 2e-16 ***

Residual standard error: 2984 on 18437 degrees of freedom
Multiple R-squared:  0.9124, Adjusted R-squared:  0.905
F-statistic: 123 on 1562 and 18437 DF, p-value: < 2.2e-16
```

```
> anova(modelA)
Analysis of Variance Table

Response: sellingprice
      Df Sum Sq Mean Sq F value    Pr(>F)
year    25  8.7687e+11  3.5075e+10 3980.3266 < 2.2e-16 ***
make    46  4.9870e+11  1.0841e+10 1230.2760 < 2.2e-16 ***
model   571  5.4783e+11  9.5943e+08 108.8768 < 2.2e-16 ***
trim    864  1.6506e+11  1.9104e+08  21.6797 < 2.2e-16 ***
body    18  7.1289e+09  3.9605e+08  44.9440 < 2.2e-16 ***
transmission  2  1.9657e+08  9.8285e+07  11.1534 1.440e-05 ***
state    37  4.6372e+09  1.2533e+08  14.2225 < 2.2e-16 ***
condition  40  3.1351e+10  7.8377e+08  88.9423 < 2.2e-16 ***
odometer   1  2.1673e+10  2.1673e+10 2459.4727 < 2.2e-16 ***
color    18  2.7395e+08  1.5220e+07   1.7271 0.02819 *
interior  16  4.1075e+08  2.5672e+07   2.9132 7.918e-05 ***
Residuals 23361 2.0586e+11  8.8121e+06
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

**Model A: Plots to check for validity of a model**



# What features are important to contribute to deciding the selling price of a used car?

**Model B: Building a model after IQR filtering and removing the outliers and comparing pvalue of different features to check for statistically significance**

```
Call:
lm(formula = sellingprice ~ factor(year) + make + model + trim +
    body + transmission + state + condition + odometer + color +
    interior, data = sample_df2)

Coefficients: (126 not defined because of singularities)

              Estimate Std. Error t value Pr(>|t|)
(Intercept)    2.467e+04  3.710e+03   6.649 3.04e-11 ***
factor(year)1991 -3.452e+03  4.206e+03  -0.821 0.411798
makeBentley      4.802e+04  2.228e+03  21.554 < 2e-16 ***

Residual standard error: 2666 on 18465 degrees of freedom
Multiple R-squared:  0.9238, Adjusted R-squared:  0.9175
F-statistic: 145.9 on 1534 and 18465 DF,  p-value: < 2.2e-16
```

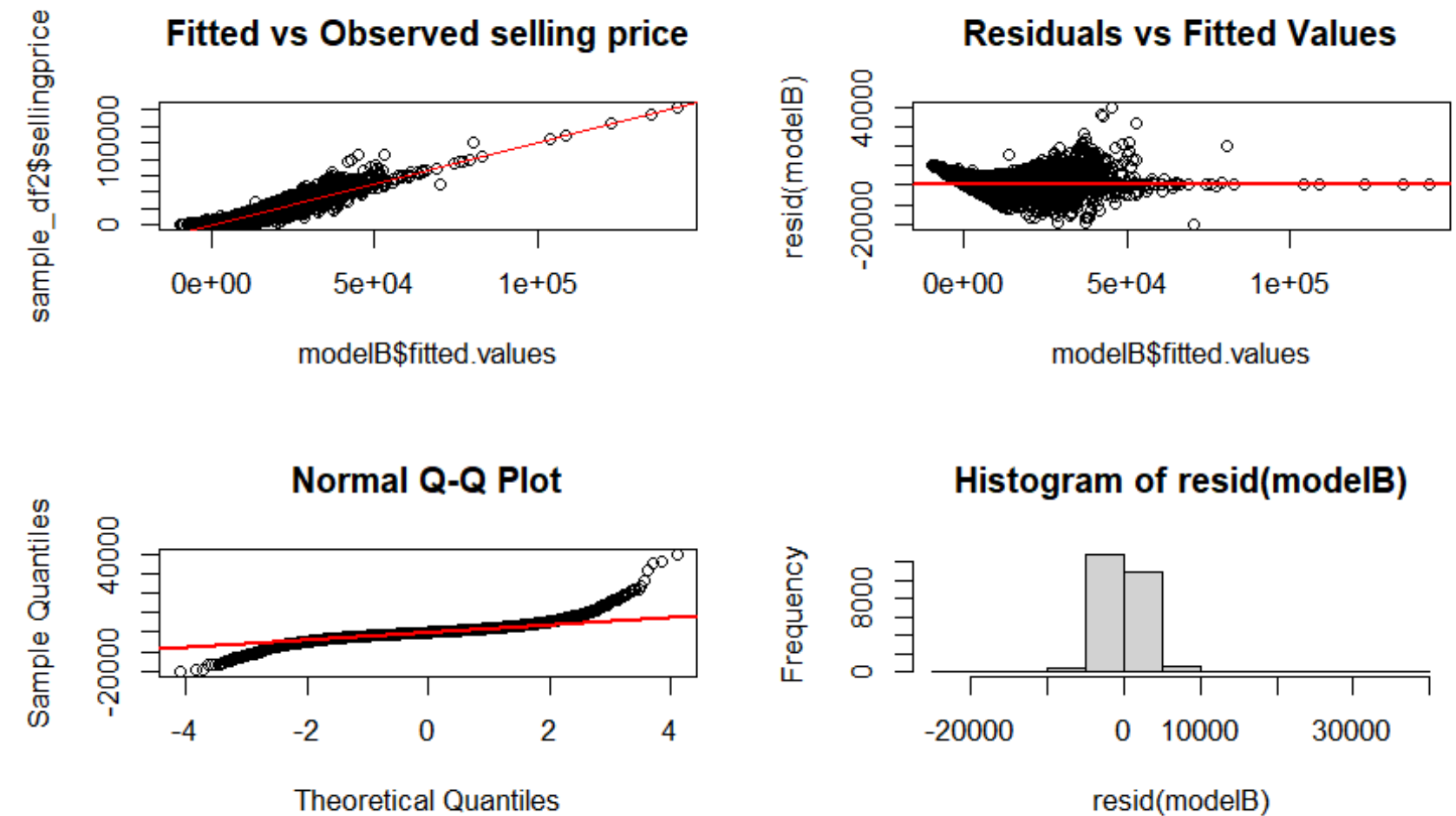
```
> anova(modelB)
Analysis of Variance Table
```

Response: sellingprice

|              | Df    | Sum Sq     | Mean Sq    | F value   | Pr(>F)    |     |
|--------------|-------|------------|------------|-----------|-----------|-----|
| factor(year) | 25    | 8.3886e+11 | 3.3554e+10 | 4878.9428 | < 2.2e-16 | *** |
| make         | 47    | 4.2727e+11 | 9.0908e+09 | 1321.8370 | < 2.2e-16 | *** |
| model        | 559   | 4.7519e+11 | 8.5006e+08 | 123.6025  | < 2.2e-16 | *** |
| trim         | 867   | 1.3753e+11 | 1.5863e+08 | 23.0648   | < 2.2e-16 | *** |
| body         | 16    | 5.5989e+09 | 3.4993e+08 | 50.8815   | < 2.2e-16 | *** |
| transmission | 2     | 3.2407e+08 | 1.6204e+08 | 23.5607   | 5.998e-11 | *** |
| state        | 37    | 2.7591e+09 | 7.4571e+07 | 10.8428   | < 2.2e-16 | *** |
| condition    | 40    | 1.9003e+10 | 4.7507e+08 | 69.0765   | < 2.2e-16 | *** |
| odometer     | 1     | 2.7586e+10 | 2.7586e+10 | 4011.0812 | < 2.2e-16 | *** |
| color        | 19    | 2.8138e+08 | 1.4810e+07 | 2.1534    | 0.002494  | **  |
| interior     | 15    | 4.2791e+08 | 2.8527e+07 | 4.1479    | 1.071e-07 | *** |
| Residuals    | 23371 | 1.6073e+11 | 6.8774e+06 |           |           |     |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**Model B: Plots to check for validity of a model**





# What features are important to contribute to deciding the selling price of a used car?

**Model C: Building a model by combining columns and grouping the categories and comparing pvalue of different features to check for statistically significance**

```
Call:
lm(formula = log(sellingprice) ~ year + make_model_trim + transmission +
    state + condition + log(odometer) + interior, data = sample_df3)

Residuals:
    Min       1Q   Median       3Q      Max
-3.13824 -0.22054 -0.00084  0.23691  2.72819

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    6.635971   0.278968  23.788 < 2e-16 ***
year1992        0.365420   0.326882   1.118 0.263623
make_model_trimChevrolet_Cruze_1LT -0.669457   0.054534 -12.276 < 2e-16 ***
interiorwhite    0.126450   0.157568   0.803 0.422266
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4435 on 24830 degrees of freedom
Multiple R-squared:  0.7592, Adjusted R-squared:  0.7576
F-statistic: 463.3 on 169 and 24830 DF, p-value: < 2.2e-16
```

```
> anova(modelC)
```

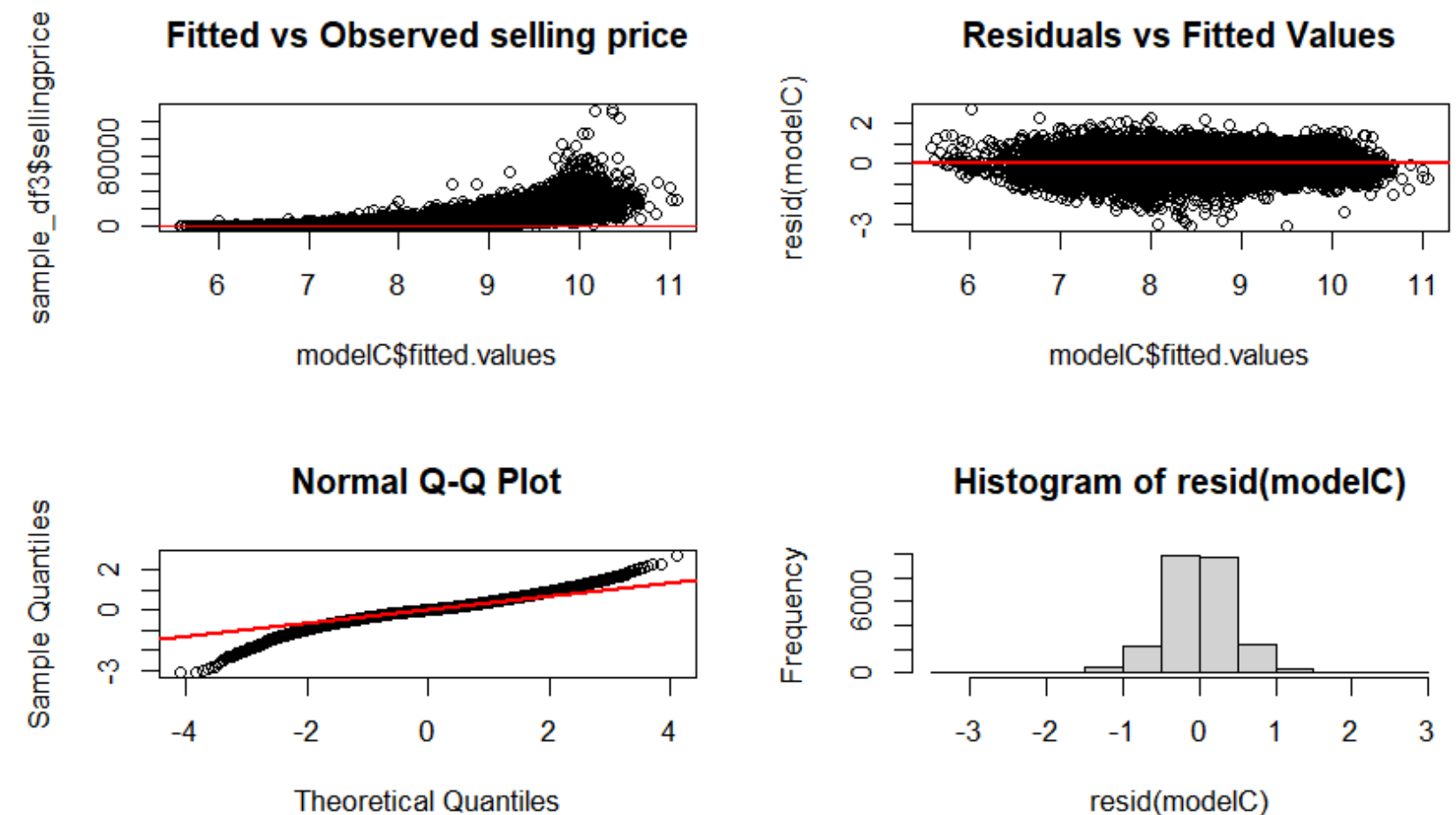
Analysis of Variance Table

Response: log(sellingprice)

|                 | Df    | Sum Sq  | Mean Sq | F value  | Pr(>F)        |
|-----------------|-------|---------|---------|----------|---------------|
| year            | 24    | 13280.6 | 553.36  | 2813.730 | < 2.2e-16 *** |
| make_model_trim | 50    | 843.4   | 16.87   | 85.767   | < 2.2e-16 *** |
| transmission    | 2     | 10.5    | 5.25    | 26.718   | 2.565e-12 *** |
| state           | 37    | 161.9   | 4.38    | 22.247   | < 2.2e-16 *** |
| condition       | 40    | 903.5   | 22.59   | 114.856  | < 2.2e-16 *** |
| log(odometer)   | 1     | 108.2   | 108.21  | 550.207  | < 2.2e-16 *** |
| interior        | 15    | 90.4    | 6.02    | 30.628   | < 2.2e-16 *** |
| Residuals       | 24830 | 4883.2  | 0.20    |          |               |

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

## Model C: Plots to check for validity of a model



# What features are important to contribute to deciding the selling price of a used car?

## **1. Vehicle Age (Year)**

- Newer cars consistently sell for higher prices across all models.

## **2. Condition Score**

- Better condition strongly increases selling price, reflecting wear and tear.

## **3. Mileage (Odometer)**

- Higher mileage reduces price; one of the most influential continuous predictors.

## **4. Make–Model–Trim (Vehicle Identity)**

- Brand and specific variant significantly affect pricing; luxury and popular models command premiums.

## **5. Market Factors (State & Transmission)**

- Regional demand differences and transmission type (automatic vs manual) both impact resale value.

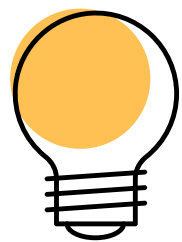


# Conclusion

- Final Analysis and Business Values
- Limitations
- Next Steps

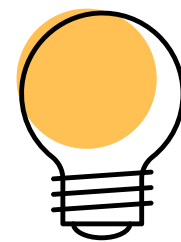
# Final Analysis Business Values

## Final Analysis



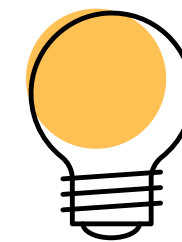
**Location** Play a Big Role in Used-Vehicle Prices.

There is a dramatic difference of average Used-vehicle prices among states.



**Vehicle Brand** Affects the Valuation of Used-Vehicles.

The brand of the car was a determining factor of used-vehicle valuation.

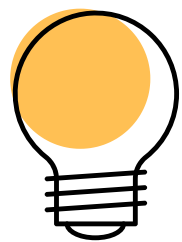


A Model's **Make Year, Trim, Transmission, state, Condition, Odometer, interior** determine the value of Used-Vehicles.

Characteristics such as year, make, model, trim, transmission, state, condition, odometer, interior determined the valuation of used-vehicles.

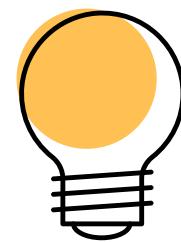
# Final Analysis Business Values

## Business Values



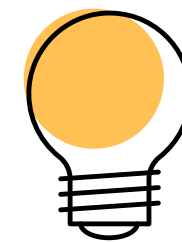
**Location** Play a Big Role in Used-Vehicle Prices.

- Sellers: Utilize state-level price deviations as a framework for pricing and inventory decisions
- Buyers: Utilize state-level price deviations to identify special markets based on location



**Vehicle Brand** Affects the Valuation of Used-Vehicles.

- Sellers: Focus more on brands that offer stronger resale value.
- Buyers: Choose high-value-retention brands when making purchases.

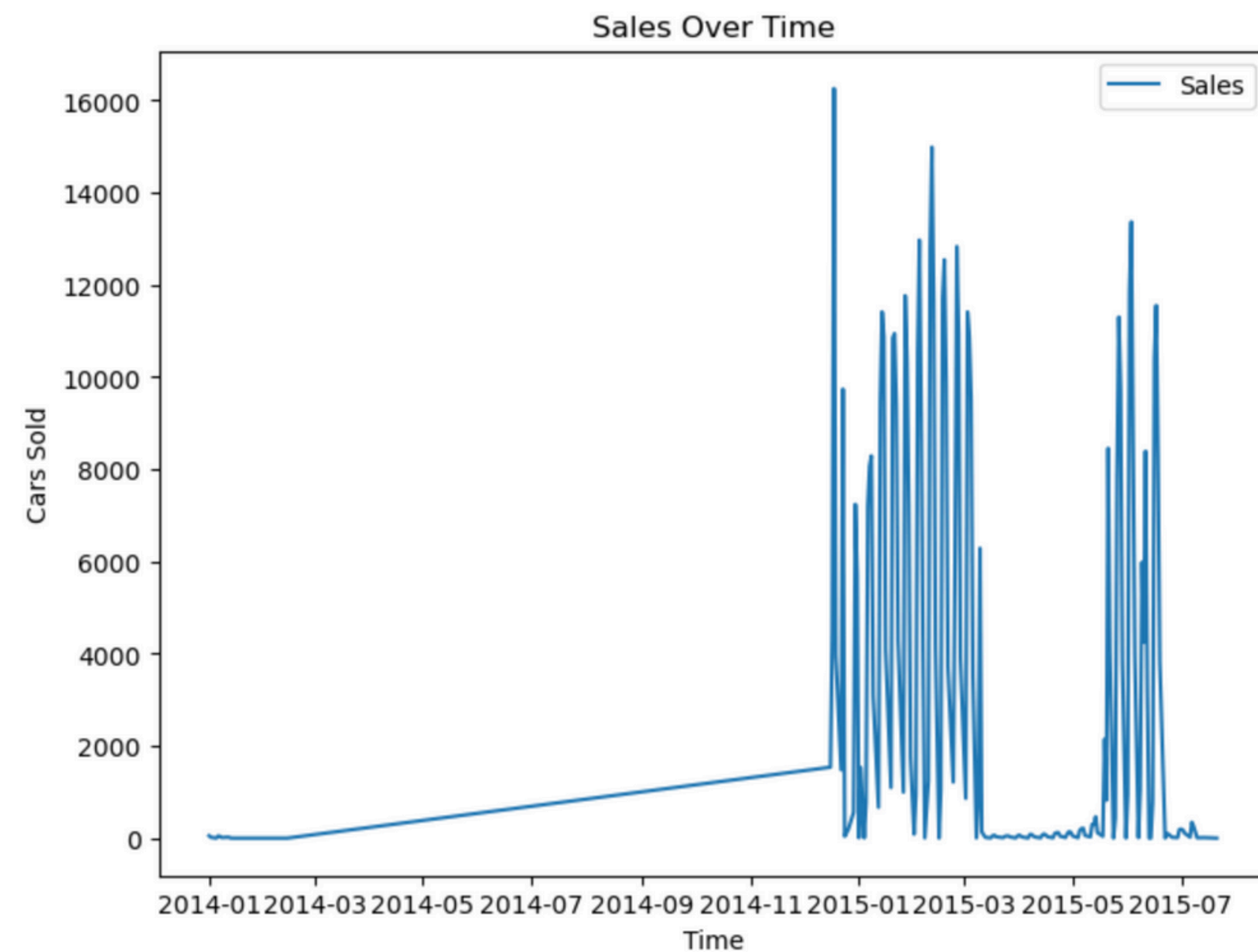


A Model's **Trim, Transmission, state, Condition, Odometer, interior** determine the value of Used-Vehicles.

- Sellers: Adjust the vehicle's price according to its characteristics
- Buyers: Evaluate the fairness of the vehicle based on such characteristics.

# Limitations

## Absence of Time Series Analysis



- Contains dates between January 2015 and July 2015 with some data from January 2014.
- Difficult to do a valid time-series analysis.
- Period is a week after we broke down the chart into two smaller charts for Jan 2015–April 2015 and May 2015 – July 2015.
- Each weekend has a significant dip which may allude to the dataset not being complete

# Next Steps

01

Expand the dataset beyond the current 8 months to capture full yearly and seasonal pricing patterns

02

Incorporate additional attributes like MSRP to measure depreciation more accurately

03

Use the repeat-sales dataset (duplicate VINs) to track how the same vehicle depreciates across multiple transactions

04

Explore predictive models like Random Forests to capture non linear relationships



Thank you!